TABLE 1: The codebook developed during our qualitative analysis. This codebook contains primary and secondary-level codings, as well as a description and an example for each code.

Code	Description	Example Quotes
Background & Responsibilities	Information related to participants such as industry experience, position, and demographic information	—
Position & Responsibilities	Describes participants' positions and their responsibilities.	I'm an associate vice president of application development.
Industry Related Experience	Information about the projects participants work on and the fields they are involved in.	I supported various projects related to electric vehicles as well as the autonomous vehicle.
Scenario-based Questions	Explores how participants describe situations where CAVs encounter unexpected roadblocks or traffic diversions.	—
Challenging Example	Describes challenging situations participants encounter with autonomous vehicles and their responses.	A vehicle that was stuck in a construction zone.
Possible Solutions	Describes how participants suggest or consider solutions to challenges faced by CAVs.	It can be mitigated through testing.
Teleoperation & Capabilities of Teleoperator	Explores participants' understanding of teleoperation and the expected roles or capabilities of a teleoperator.	—
Definition of Teleoperation	Captures how participants define or describe the concept of teleoperation.	The concept of operation is that if a self-driving car encounters an accident, there should be someone behind the screen who can take control of the vehicle.
Experience with Teleoperation	Describes participants' direct or indirect experiences with teleoperation systems.	If the robot is tripped or falls over, the remote operator has full access to all sensors, such as cameras and LiDAR, to view the surroundings.
Importance of Teleoperation	Participants expressed how important they believe teleoperation is.	Teleoperation is very necessary for CAVs
Use Case of Teleoperation	Explain the situations in which teleoperation is used.	The algorithm is not working, or some of the algorithm is stuck, and the malfunction happens during driving.
Teleoperation vs. CAV	The key differences between teleoperated vehicles and fully autonomous vehicles.	For a fully autonomous vehicle, we assume that the controller is implemented on the vehicle locally, while a teleoperated vehicle, either remotely, by a human or by a computer, whichever.
Passengers' Expectations of a Teleoperator	What passengers expect from a teleoperator, like their desired level of ability and responsiveness.	I would expect they minimally had some type of chauffeur's license, some type of qualification saying they were qualified to take other people as passengers.
Teleoperating a Fleet of RoboTaxi or Trucks	The use of teleoperation for vehicle fleets.	I think most of the Baidu or what's the company that's in China, Xiaopeng vehicle, or it's probably Xiaopeng.
Teleoperating Personal Vehicles	The use of teleoperation for private vehicles.	
Teleoperator Monitoring CAV	How teleoperators monitor CAVs, like data they access, and how they determine when to intervene.	vehicle should sound an alarm and that's when the teleoperator interferes
Teleoperation Training, Certification & Licensing	A teleoperator is required to have specific skills to remotely operate a CAV.	For remote drivers, we need them to have CDL license.
Teleoperator Access	Describes the types of system access teleoperators have, such as control over vehicle functions and access to real-time sensor data.	Teleoperator should have access to or should monitor the vehicle's status.
Teleoperator Service	Gain insight into the experts' expectations for teleoperation as a service.	I think if teleoperation was something that folks could subscribe to as a service on their vehicle, then I would expect that there's some level of oversight that they're providing.
Teleoperator Identity	Explores the identity of the teleoperator, including their role, background, and whether passengers or stakeholders are aware of who is remotely controlling the vehicle.	if the teleoperator goes through stringent background checks, then we have at least made sure that whoever is given the authorization to control the vehicle.
Security and Privacy Challenges	Explores the security and privacy challenges associated with CAVs and teleoperation.	—
Security Concerns	Describe security concerns related to teleoperation and CAVs.	I think the main thing is signal jamming.
Malicious Teleoperator	Describe the risks posed by a malicious teleoperator, including intentional misuse of vehicle control, data breaches, and the potential for compromising passenger safety and system integrity.	If we make this assumption, we cannot predict what the impact will be on robot taxi providers, because robot taxi providers have teleoperators.
Reporting Security Incidents	Describe how security incidents in teleoperation and CAV systems are reported.	They don't even report it because it's a bad history.
Importance of Security in Design and Implementation	Discussion on why integrating security into the design and implementation phases is important, and how it affects system reliability and trustworthiness.	We are looking at some cybersecurity issues on the ELD devices.
Security Testing	Discussion on the role of security testing in ensuring teleoperation system robustness.	You just do matrix multiplication, then it can be done.
Organizational Security & Safety Training	Discussion on how organizations provide security and safety training for teleoperators and related staff.	If you are actually doing the work for um electric uh electric vehicles there are also advanced safety training now.
User Privacy Violations	Discussion on incidents or concerns related to violations of user privacy in teleoperation and CAV systems.	If this could be really a privacy issues could be dangerous because if you say something that nobody should hear.
Privacy Concerns	Discussion on concerns related to the collection, use, and storage of personal data.	So maybe also outside as well, you're going to take in some of the other people's, the car plate or some other information, which is also very sensitive.
Data Transfer Mechanism	Discussion on how data is transmitted between the vehicle and the edge server or cloud.	I believe the most popular way is to send it from mobile internet, like 4G or 5G internet.
Uses of Collected Data	Describe the data usage.	Big data to train the big model.
Necessary Data Collected	Explore which types of data are necessary to collect.	They collect LiDAR data, camera data, radar data, IMU data, and GPS data.
Data Protection in Transit	Discussion on how data is protected during transmission between vehicles and the edgeserver/cloud.	Common practices include, encryption and decryption.
Data Protection in Vehicles	Discussion on how data is stored and protected within the vehicle.	I think we also have some teams try to protect the data itself.

Code	Description	Example Quotes
Security and Safety Processes in Design and Development	Discussion on the processes integrated into the design and development phases to ensure security and safety.	—
Alarming the Algorithm & Overriding the Teleoperator	Discussion on scenarios where the system detects unsafe or abnormal behavior and triggers alarms to the algorithm or overrides teleoperator commands to maintain safety.	There is an advanced function right now in the autonomous vehicle. It's called, I would say, active security. The vehicle itself can detect the road boundary.
Alarming the Teleoperator & Overriding the Algorithm	Discussion on situations where the system alerts the teleoperator about critical events and allows human intervention to override algorithmic decisions to ensure safety.	If the robot was tripped, the remote controller has full access to all the sensors, for example, the cameras and LIDARs to view the surroundings.
Passenger Security and Safety Guarantees	Discussion on how security and safety are ensured for passengers in teleoperated and autonomous vehicles.	First is the authentication, you have to authenticate both the operator and the owner of the autonomous vehicles to establish a secure and authenticated communication channel.
Possible Methods to Mitigate Safety Problems	Discussion on potential methods and strategies to address or reduce safety issues.	For steering systems we introduce some potential faults in the steering system but then we also have a stop uh stop button.
Communication concerns	Discussion on concerns related to communication between vehicles, teleoperators, and infrastructure.	You have to ensure the communication is smooth, and also there's no delay or packet loss in the communication channel.
Possible Solution for Network/Latency	Discussion on potential solutions to address network instability and latency issues in teleoperation systems.	Like switching from DSRC to 5G
Design Challenges	Discussion on the challenges faced during the design of teleoperation and CAV systems.	Need to maintain the security, safety, and comfort of the passengers.
Unusual Scenarios during Testing	Describe unexpected or rare situations encountered during testing.	The real real world data set is not what you expected.
Testing Procedures	Discussion on the procedures used to test the system.	We do a lot of simulation.
Tools and Techniques	Discussion on the tools and techniques used in the system.	We use Unreal Engine to develop the simulation.
Regulatory Guidelines and Legislation	Discussion on the laws, regulations, and guidelines that govern teleoperation, including compliance requirements, policy gaps, and regional differences.	—
Regulatory Guidelines	Discussion on official guidelines and standards that teleoperation should follow.	I don't think there is a mature law for the teleoperations.
Liability & Damages	Discussion on who is held responsible in the event of an incident involving teleoperation.	It should be the company.
Suggested Regulations	Discussion on participants' suggestions for new or improved regulations to govern teleoperation.	You have to ensure the operator is licensed and they have rich experience, and it's trustable.
Special Vehicle Skills and License	Describe the specific skills and licenses required for teleoperators or drivers to safely control CAVs.	Teleoperator can have some soft skills to make sure that the people can calm down and really follow the guidelines.
Knowledge of Regulations	Discussion on existing regulations related to teleoperation.	According to FMCSA right now.
Data Collection Regulations	Discussion on the regulations governing the collection of data in teleoperation and CAV systems.	That's some of the common sense when we're dealing with the data, we need to, uh, understand that.
Interacting with the Authorities	Discussion on how teleoperators interact with authorities.	I am not aware.
Deadlines and Pressure	Discussion on how time constraints and performance pressure affect teleoperators or developers working on the area.	There is always pressure, uh, for us to speed up.